scientific reports

Check for updates

OPEN

A privacy-aware healthcare framework with model pattern-deviation detection for heart-disease prediction using L2-GNAE and PDDP

Rashmi Dwivedi[1], Basant Kumar[2], Vivek Mishra[1], K. Hemachandran[3], Vidushi Mishra[4] & SeongKi Kim[5]✉

Sensitive patient data protection is essential to ensure medical reliability in healthcare. Yet, the traditional studies didn't analyze the deviation in the shared model pattern, thus resulting in poor diagnosis. Therefore, this article proposes a privacy-aware healthcare framework with model pattern deviation detection for Heart Disease (HD) prediction using L2 Gini Norm Auto-Encoder (L2-GNAE) and Polynomial Differential Decay Privacy (PDDP). Firstly, the patients are registered into the healthcare applications, followed by data sensing, data encryption, and hash code generation. Meanwhile, to authenticate the data integrity, the data decryption and hash code verification are done. During testing, the verified data is subjected to the trained proposed local model for HD prediction. Next, to perform model privacy, PDDP is used. Afterward, to effectively classify the HD, the Triple Gated Lipschitz Recurrent Unit (TGLRU) is utilized. Also, the local model gradients are updated in the global model, where L2-GNAE is utilized to detect the deviations in the shared model pattern. If the deviation is detected, then the alert is sent to the hospital; otherwise, the model update is carried out. The experimental testing of the proposed framework is done by using the "Heart Disease Prediction Dataset". From the validation, the proposed framework achieves 99.2145% accuracy, 99.0237% precision, and 99.1046% F-measure during HD prediction. Thus, the proposed work significantly outperforms the traditional works by obtaining a high security level (256 bits) with enhanced privacy-preserved HD prediction in healthcare maintenance.

Keywords Privacy-preservation (PP), Sensitive healthcare data (SHD), Shared model pattern deviation (SMPD), Heart disease (HD), Artificial intelligence (AI)

Globally, HD remains one of the leading causes of mortality[1]. HD is a general term for a group of conditions that affect the structure and function of the heart[2]. These conditions include coronary artery disease, heart failure, arrhythmias (irregular heartbeats), congenital heart defects, and heart valve disorders[3]. Also, HD can reduce the heart's ability to pump blood effectively, leading to serious health complications, such as heart attack, stroke, or cardiac arrest[4]. Thus, to save the patient's life, early prediction is fundamental[5]. Nevertheless, HD prediction models heavily depend on sensitive patient information[6,7]. The use of sensitive data significantly increases privacy concerns[8]. Therefore, a privacy-preserving healthcare framework has emerged as a potential solution[9]. Also, Federated Learning (FL) plays an important role in enabling collaborative model training across multiple healthcare institutions without sharing raw patient data, thereby enhancing data privacy and regulatory compliance[10,11].

Besides, the existing studies integrated Recurrent Neural Network (RNN), Convolutional Neural Network (CNN), and Gated Recurrent Unit (GRU) with FL to perform privacy-aware HD diagnosis[12,13]. These techniques excel in predicting HD by effectively learning complex nonlinear relationships from clinical data[14]. However, they

[1]Department of Pure and Applied Mathematics, Alliance University, Bangalore, India. [2]Modern College of Business and Science, Muscat, Oman. [3]AI Research Centre, Woxsen University, Hyderabad, India. [4]Christ University, Bengaluru, India. [5]Department of Computer Engineering, Chosun University, Gwangju 61452, South Korea. ✉email: skkim@chosun.ac.kr

often suffer from privacy leakage, limited interpretability, and vulnerability during collaborative learning[15,16]. Also, some prevailing works suffered from data leakage risk, insider threats, malicious model updates, and unauthorized data manipulation, leading to limited security[17]. But, the traditional studies didn't concentrate on analyzing the deviation in the shared model patterns. Thus, this research designs a privacy-aware healthcare system with SMPD analysis for HD prediction.

Research overview

This section clearly presents the existing drawbacks and the purpose of the proposed work.

Research gap The prevailing studies didn't concentrate on analyzing the deviation in the shared model pattern during healthcare system design, thus resulting in improper diagnosis.

Problem statement

The core problems of the existing works are given below,

- The conventional framework[18] failed to protect the SHD during communication, increasing data leakage risk.
- The traditional approach[19] didn't address the insider threats, where the input data was slightly modified, thus resulting in high misclassification.
- Some of the existing studies lacked transparency, limiting personalized treatment decisions.

Significance and scope of the study The proposed approach is designed to analyze the deviations in the shared model pattern, thereby mitigating inaccurate diagnoses. Further, the research methodology significantly contributes to protecting the SHD via model privacy.

Research aim and objectives The proposed work's primary aim is to implement a privacy-aware healthcare framework while handling SMPD. Likewise, the proposed work's key objectives are listed below,

- To analyze the SMPD, the proposed work employs L2-GNAE.
- To protect the SHD, a robust Siegel Modular Curve Cryptography (SMCC) is established.
- To mitigate the risk of insider manipulation, the proposed Murmur Quadratic Residue Hashing (MQRH)-based hash verification is done.
- To interpret the diagnosis results, an effective SHapley Kaniadakis Additive exPlanations (SHKAP) is introduced.

Novel contribution of the proposed work The proposed work introduces a novel privacy-aware healthcare framework for HD prediction with an FL mechanism. Unlike traditional FL and privacy-aware HD prediction models, the proposed framework develops a model for analyzing the deviation in the shared model pattern during healthcare system designs. However, the traditional works failed to analyze such shared model pattern deviations during HD prediction, resulting in inaccurate and inefficient diagnosis. The inclusion of model pattern-deviation detection using the proposed L2-GNAE enables the early identification of anomalous or malicious model updates. Additionally, most of the prevailing works failed to protect the SHD, detect insider threats, and provide interpretability during HD prediction through the FL mechanism, resulting in data leakage, misclassification, and limited personalized decision-making. Therefore, by introducing the SMCC, MQRH, and SHKAP, the overall effectiveness of the proposed framework is enhanced, thereby ensuring secure, robust, and transparent communication during HD prediction.

The paper is drafted as: "Related survey" section presents the literature review, "Proposed methodology for privacy-aware HD prediction framework" section depicts the proposed work's mathematical overview, "Results and discussion" section validates the proposed methodology's performance, and "Conclusion" section concludes the paper.

Related survey

Torfi et al.[18] applied a differentially private synthetic medical data generation framework based on a convolution generative adversarial network. Here, while ensuring privacy, Rényi differential privacy was employed to synthetically generate medical data. Yet, this model failed to encrypt the patient information, increasing security risks.

Lian et al.[20] offered an efficient and privacy-enhanced federated edge learning for healthcare cyber-physical systems. Here, the local DP technique was utilized to protect the SHD. The inclusion of ring topology communication mitigated the centralization errors. However, this model had a risk of congestion owing to the ring-based network architecture.

Fang et al.[21] recommended a differentially private medical data generation framework using a conditional tabular generative adversarial network. This framework maintained privacy while ensuring the generated data's quality. Nevertheless, owing to the minmax game theory, this model had improper privacy loss estimation, degrading the healthcare reliability.

Tian et al.[22] established a privacy-preserving decentralized deep Federated Learning (FL) training framework for securing digital healthcare. Here, to enhance the communication efficiency, ring FL and ring allreduce-centric data sharing schemes were established. Nevertheless, owing to the class imbalance issue, this framework had suboptimal performance.

Pandey et al.[19] established a robust FL scheme with Bullyan-based adaptive differential privacy framework for malicious behavior mitigation. The Bullyan-based aggregator eliminated outlier gradients to deactivate adversarial behaviors. This model had high flexibility; however, it didn't address the risk of insider threats.

Alammar et al.[23] presented a Federated eXplainable AI (FedXAI) framework for privacy-preserving cardiovascular monitoring using consumer electronics. Here, this model integrated FL with explainable AI

methods, such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), and rule-based models, to provide interpretable predictions for clinicians. As per the result, this model achieved 90% diagnostic accuracy. However, this model suffered from high computational overhead due to the complex architecture.

Shajimon et al.[24] implemented an Internet of Things (IoT)-based Vertical Federated Learning (VFL) framework for heterogeneous data-driven cardiovascular disease risk prediction. Here, this model trained the distributed neural network across multiple clients with disparate feature sets for predicting cardiovascular disease accuracy. As per the validation, this model achieved 90% predication accuracy. Yet, this model failed to ensure data privacy due to the absence of complete data access at many participating clients.

Naveenkumar et al.[25] introduced an FL-based hybrid deep learning transformer model for HD detection with XAI. Here, the HD detection was carried out by combining a sparse attention-assisted convolutional DenseNet transformer with differential privacy and SHAP-based interpretability. As per the experimental validation, this model achieved superior performance and transparency with improved accuracy. Yet, this model suffered from limited scalability and robustness due to its ability to process only a limited number of data samples.

Proposed methodology for privacy-aware HD prediction framework

The proposed work designs a privacy-aware healthcare framework with SMPD detection for HD prediction. In Fig. 1, the proposed work's architecture is shown.

Patient registration

Initially, the patients are registered using their username, password, and personal details.

$$S_p = \{S_1, S_2, \ldots\ldots S_P\} \text{ where, } \quad p = 1 \text{ to } P \tag{1}$$

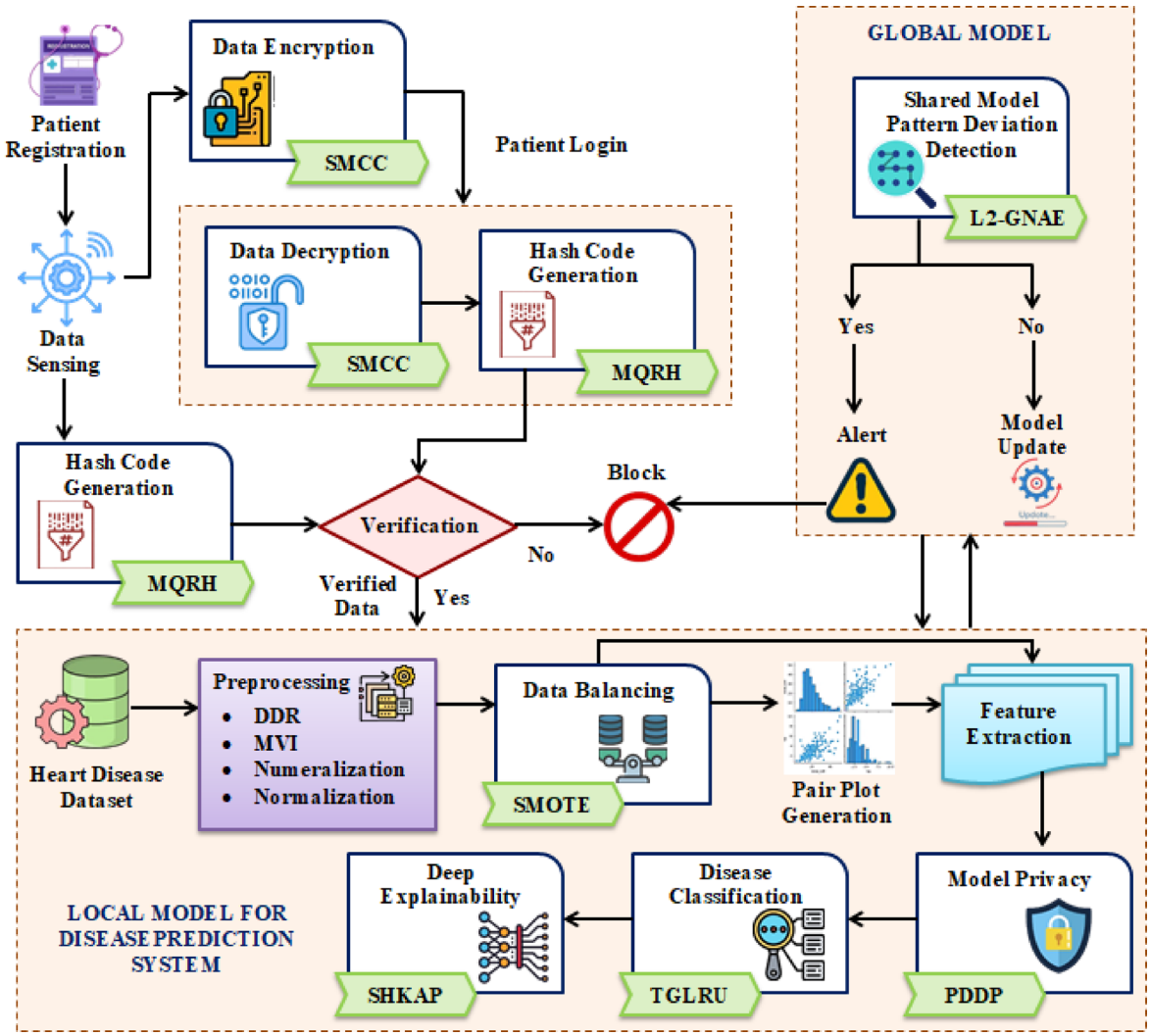


Fig. 1. Conceptual design of the proposed methodology.

Here, P displays the number of registered patients S_p. Also, the security keys are generated for each patient using SMCC. From the patients, the healthcare data (D_∞) is sensed to perform further healthcare diagnosis.

Data encryption

Here, to encrypt the D_∞, the proposed SMCC is utilized, thus protecting the SHD. The Elliptic Curve Cryptography (ECC) has a high security level. Yet, the ECC is sensitive to the choice of elliptic curve. Therefore, the proposed work employs the Siegel Modular Curve (SMC)(C) to ensure unified patterns in computation.

$$C = X\left(2h, \mathrm{Z}\right) \backslash U_{\mathrm{space}} \tag{2}$$

Here, h depicts the genus of a curve, Z denotes a period matrix, U_{space} illustrates the Siegel upper half-space, and X implies the symplectic modular group. Next, the private key (a) and public key (b) are generated, followed by ciphertext (h_1, h_2) creation.

$$b = a \cdot C\left(j\right) \tag{3}$$

$$h_1 = \mathrm{rnd} * C\left(j\right) \tag{4}$$

$$h_2 = D_\infty + rnd \times b \tag{5}$$

where, $C\left(j\right)$ denotes the curve parameter and rnd illustrates the random value. Then, the data decryption is done to obtain the encrypted data (h_v).

$$D_\infty = h_2 - a * h_1 \tag{6}$$

$$h_v = (h_1, h_2, \ldots\ldots h_V) \ \text{ Where, } v = 1, 2, \ldots V \tag{7}$$

Here, V indicates the number of h_v.

Hash code generation

Now, by using the MQRH algorithm, the hash code of D_∞ is generated for insider threat mitigation. The Murmur Hashing (MH) algorithm has a low collision rate. However, the MH approach has poor randomness due to the improper seed value initialization. Therefore, the proposed method utilizes the Quadratic Residue Equation (QRE) to determine the appropriate seed value with high randomness.

Firstly, D_∞ is divided into 32-bit blocks (N_w), followed by block updation.

$$N_w = [N_1, N_2, \ldots\ldots N_W] \ \text{ Here } \ w = 1 \,\text{to}\, W \tag{8}$$

$$N_w^{\otimes} = N_w \times u_1 \tag{9}$$

$$N_w^{\Omega} = \mathrm{Rt}\left(N_w^{\otimes}, u_2\right) \tag{10}$$

$$N_w^{\mathrm{mix}} = N_w^{\Omega} \times u_3 \tag{11}$$

Here, Rt depicts the rotation shift operation, $N_w^{\otimes}$, N_w^{Ω}, and N_w^{mix} illustrate the updated blocks after multiplication, rotation, and mixing, respectively, and $(u_1, u_2, u_3, u_4, u_5, u_6)$ demonstrates the fixed constants. Subsequently, the hash value is updated based on the updated blocks. Here, the seed value $\left(\ddot{I}\right)$ is determined using the QRE.

$$\ddot{I} = N_w^2 \bmod \mathrm{pi} \tag{12}$$

Here, pi demonstrates the prime number.

$$I' = \ddot{I} \oplus N_w^{\mathrm{mix}} \tag{13}$$

$$I'' = \mathrm{Rt}\left(I', u_4\right) \tag{14}$$

$$I''' = I'' \times u_5 + u_6 \tag{15}$$

where,I',I'', and I''' denote the updated hash values at each mixing round. Here, the final hash value is depicted as I'''.

In real-time, the patients log into the system and then provide their data (Dt). Meanwhile, the encrypted patient's data is decrypted, followed by hash generation. Next, the generated hash value is compared with the hash value created during registration to verify the data integrity. Next, the verified data $(\mathrm{Dt}_{\mathrm{verify}})$ is fed into the proposed local model for HD prediction.

Local model for the HD prediction system

To predict HD, the proposed work trains a local model further.

HD dataset

Initially, the historical HD detection dataset $\left(\hat{T}\right)$ is gathered to train the local model.

Pre-processing Next, to enhance the data quality, the $\hat{T}$ is pre-processed. Primarily, the duplicated data in $\hat{T}$ is eliminated. Then, de-duplicated data is subjected to the Missing Value Imputation (MVI), where the missing values are imputed by considering the non-missing values' mean value, followed by numeralization. Subsequently, the numeralized data (Xy) is normalized within the range of 0–1 utilizing min–max normalization.

$$\tilde{O} = \frac{\mathrm{Xy} \text{ - } \min(\mathrm{Xy})}{\max(\mathrm{Xy}) - \min(\mathrm{Xy})} \tag{16}$$

Here, $\tilde{O}$ specifies the pre-processed data.

Data balancing The proposed work utilizes the Synthetic Minority Over-sampling Technique (SMOTE) to perform data balancing in $\tilde{O}$. Initially, the minority class is chosen; then, instances are selected randomly from the minority class. Next, the k-nearest neighbours are detected. From the k-nearest neighbour, a random neighbour is chosen. Based on the selected instance (Si) and the chosen neighbour (Sn), the new samples (Nw) are generated as follows,

$$\mathrm{Nw} = \mathrm{Si} + \overleftrightarrow{\mathrm{R}} * (\mathrm{Sn} \text{ - } \mathrm{Si}) \tag{17}$$

Here, $\overleftrightarrow{\mathrm{R}}$ indicates the random number. Thus, the above steps are repeated until the balanced data (B_{∞}) is obtained.

Pair plot generation From B_{∞}, the pair plot is generated to visualize the linear and non-linear relationships between pairs. A pair plot $\left(\tilde{A}\right)$ is created by plotting every numerical feature against every other feature in a grid of scatter plots.

Feature extraction Here, the features like age, sex, chest pain type, resting blood pressure, serum cholesterol level, and number of major vessels are extracted from B_{∞}. Moreover, the pair plot features, such as trends, outliers, and correlation, are extracted. Thus, the extracted features $\left(\overleftrightarrow{L}\right)$ are subjected to the model privacy.

Model privacy Model privacy is applied in $\overleftrightarrow{L}$ using the proposed PDDP to protect the SHD. When compared with other traditional methods like Synthetic Data Generation (SDG), K-Anonymity, L-Diversity, Local DP (LDP), Central DP (CDP), and Stochastic DP (SDP), the DP effectively reduces the risk of information leakage by providing provable, mathematical guarantees against re-identification and linkage attacks. Nevertheless, the DP has suboptimal performance due to the trade-off issue betwixt privacy and accuracy. Therefore, the privacy budget is selected using the Polynomial Decay Function (PDF) that dynamically adjusts the privacy budget by actively modulating the noise injected during model training. Unlike conventional differential privacy methods, the proposed PDDP dynamically adjusts the privacy budget using a PDF. This illustrates that the value of epsilon gradually decreases according to a polynomial schedule as training progresses. This adaptive decay strengthens privacy protection over time by increasingly limiting the sensitivity of model updates. Also, the decay mechanism provides a controlled trade-off between privacy preservation and predictive performance.

Firstly, the privacy parameter, such as epsilon $(\mathrm{eps_v})$, is determined using the PDF.

$$\mathrm{eps}_{\hat{\mathrm{v}}} = \frac{\mathrm{eps}_0}{(1 + \mathrm{D_{rate}}\hat{\mathrm{v}})^{\mathrm{Pw}}} \tag{18}$$

Here, eps_0 specifies the initial privacy budget, $\mathrm{D_{rate}}$ illustrates the decay rate, $\hat{\mathrm{v}}$ denotes the iteration, and Pw demonstrates the polynomial power.

Also, the sensitivity $\left(\hat{K}\right)$ is measured to showcase how much the query output can change if one individual's data is modified. Next, the Gaussian mechanism is applied to inject the noise in $\overleftrightarrow{L}$.

$$O_{\mathrm{noisy}} = \frac{\hat{K}\overleftrightarrow{L}}{\mathrm{eps}_{\hat{\mathrm{v}}}} \tag{19}$$

where, O_{noisy} illustrates the privacy-preserved feature values.

Disease classification The proposed TGLRU effectively performs HD classification based on the O_{noisy}. The Gated Recurrent Unit (GRU) has less memory requirement. But, the GRU has over-fitting issues and low learning efficiency. Therefore, the proposed work introduces the Lipschitz Regularization (LR) technique to regularize the weight coefficients, minimizing overfitting issues. To augment the neurons' learning efficiency, the proposed approach deploys the Triple Activation (TA) function that introduces non-linearity. In Fig. 2, the structure of TGLRU is given.

Primarily, the input O_{noisy} and previous hidden state (Hn_{o-1}) are transmitted to the reset gate (i_{reset}) that determines the amount of forgotten information.

$$i_{\mathrm{reset}} = \mathrm{Trip} \cdot \left(O_{\mathrm{noisy}}, \mathrm{Hn}_{o-1}\right) \times f_{weight} + B_{\mathrm{ias}} \tag{20}$$

$$\mathrm{Trip}\left(O_{\mathrm{noisy}}\right) = \mathrm{ax} \cdot O_{\mathrm{noisy}}^{3} \tag{21}$$

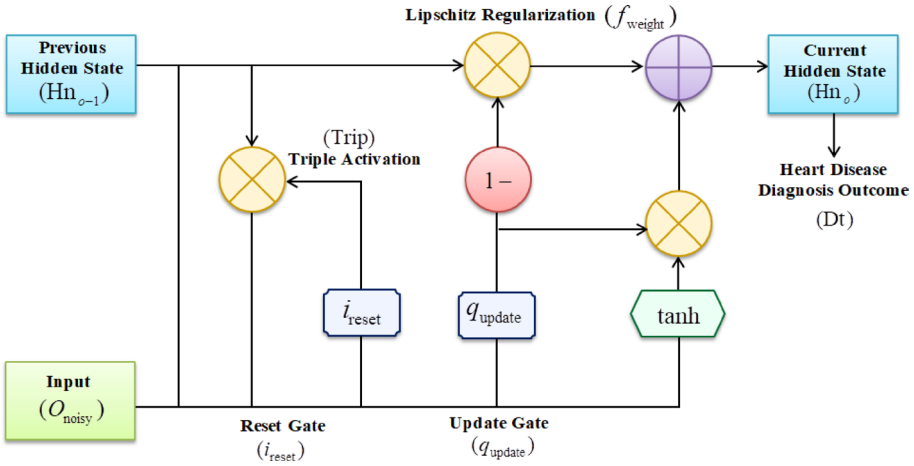


Fig. 2. Architecture of TGLRU.

where, Trip denotes the TA function, ax indicates the trainable parameter, f_{weight} depicts the weight value, and B_{ias} illustrates the bias value. Here, the f_{weight} is regularized using the LR.

$$f_{weight} \Rightarrow \|L_{fun}\left(O_{\text{noisy}}\right) - L_{fun}\left(\text{Hn}_{o-1}\right)\| \leq C_{\text{nt}} \|O_{\text{noisy}} - \text{Hn}_{o-1}\| \tag{22}$$

Here, L_{fun} denotes the Lipschitz continuous function and C_{nt} indicates the Lipschitz constant. Then, the update gate (q_{update}) is computed to determine the amount of relevant information.

$$q_{\text{update}} = \text{Trip} \times \left(\left(O_{\text{noisy}}, \text{Hn}_{o-1}\right) \cdot f_{weight}\right) + B_{\text{ias}} \tag{23}$$

Next, the current hidden state (Hn_o) is determined to obtain the class probability.

$$\text{Hn}_o = (1 - q_{\text{update}}) * \text{Hn}_{o-1} + q_{\text{update}} * \ddot{\text{H}}\ddot{\text{n}}_o \tag{24}$$

Here, $\ddot{\text{H}}\ddot{\text{n}}_o$ denotes the candidate hidden state. Lastly, the class with the highest probability is considered as the detection outcome (Dt). The pseudocode of TGLRU is illustrated below.

Input: Privacy-preserved feature values O_{noisy}
Output: HD detection Dt
Begin
  Initialize O_{noisy}, Hn_{o-1}, f_{weight}, Trip and B_{ias}
  For 1 to each input do,
    Compute reset gate
      $i_{\text{reset}} = \text{Trip} \cdot \left(O_{\text{noisy}}, \text{Hn}_{o-1}\right) \times f_{\text{weight}} + B_{\text{ias}}$
    Activate TA function Trip
      $\text{Trip}\left(O_{\text{noisy}}\right) = \text{ax} \cdot O_{\text{noisy}}^{\ 3}$
    Regularize weight value f_{weight} using LR
    Determine update gate q_{update}
    Compute hidden state
      $\text{Hn}_o = \left(1 - q_{\text{update}}\right) * \text{Hn}_{o-1} + q_{\text{update}} * \ddot{\text{H}}\ddot{\text{n}}_o$
    If (actual == target) {
      Terminate
    Else
      Back − propagation }
    End If
  End For
  Return Dt
End

Deep explainability

Here, to provide detailed insight into Dt, the proposed SHKAP is utilized centered on the O_{noisy}. The SHapley Additive exPlanations (SHAP) has better consistency. However, the SHAP is sensitive to the choice of weight

value. Hence, the proposed work employs the Kaniadakis Entropy (KE) function to assign the weight value. The Shapley values are weighted to visualize the feature importance using the KE function (K_{ent}).

$$K_{\text{ent}} = \sum_{u'''} O_{\text{noisy}}^{u'''} \ln_{\text{ix}} \left(O_{\text{noisy}}^{u'''} \right) \tag{25}$$

where, ix exhibits the entropic index and $u''' = 1 \text{ to } U'''$ indicates the number of features in the feature set.

$$J_{\text{shap}} = \sum_{\dddot{z} \subset O_{\text{noisy}}} \frac{|\dddot{z}|! \left(|O_{\text{noisy}}| - |\dddot{z}| - 1\right)!}{|\dddot{z}|!} \left[O_{\text{noisy}} \left(\dddot{z} \cup U''' \right) - \left(O_{\text{noisy}} \right) \right] * K_{\text{ent}} \tag{26}$$

Here, J_{shap} demonstrates the weighted Shapley values and $\dddot{z}$ indicates the subset of features. Based on J_{shap}, the O_{noisy} is effectively interpreted, thus enhancing the model's trust.

Global model

A global model is a centralized model created by fusing updates $(\text{U}\hat{\text{p}})$ from multiple local models.

Model pattern deviation detection

To perform SMPD detection, the proposed work establishes the L2-GNAE. The existing Auto-Encoder (AE) automatically learns patterns without labels compared to the other traditional methods like Variational AE (VAE), Denoising AE (DAE), and Sparse AE (SAE). These traditional methods rely on additional structural constraints or stochastic assumptions that may limit their flexibility in model pattern deviation analysis. However, the AE has considerable computational overhead, as it treats all feature contributions uniformly and struggles to handle high-dimensional features. Therefore, the proposed work employs the L2 Gini Norm Filtering (L2GNF) technique. The L2GNF reduces complexity by identifying and filtering less informative or highly repetitive feature components while preserving the most discriminative characteristics of model updates. Thus, L2-GNAE learns the normal behavioral pattern of federated model updates in a compressed latent space and then identifies deviations by considering the learned patterns. Further, the model patterns' deviations are accurately detected based on the reconstruction loss generated by the L2-GNAE, ensuring effective identification of subtle and long-term variations in shared model behavior. Higher reconstruction loss indicates the presence of model pattern deviations, while the lower reconstruction loss signifies that the incoming model updates follow the normal behaviour.

Primarily, the encoder $\left(E^{\text{coder}}\right)$ applies weighted transformations (W_{form}) and L2GNF (L_{GNF}) to learn the most important features of normal patterns.

$$L_{\text{GNF}} = \sqrt{\sum_{t=1}^{T} \text{U}\hat{\text{p}}_t^2} \times \frac{\sum_{t=1}^{T} |\text{U}\hat{\text{p}}_1 - \text{U}\hat{\text{p}}_t|}{2T^2 \tilde{M}} \tag{27}$$

$$E^{\text{coder}} = W_{\text{form}} \cdot \left(L_{\text{GNF}}\right) \tag{28}$$

where,$\tilde{M}$ indicates the mean vector of the $\text{U}\hat{\text{p}}$ and $t = 1\, toT$ illustrates the number of sample patterns. Then, the bottleneck layer (B_{neck}) is used to capture a compact shared structural pattern $(L_{pattern})$ of normal data.

$$B_{neck}\left(\text{U}\hat{\text{p}}\right) \xrightarrow{\text{capture}} L_{pattern} \tag{29}$$

Finally, the decoder reconstructs the original input from the latent representation to compute the error value. Based on the error value, the deviation (Dv) in the shared model pattern is analyzed. The pseudocode of L2-GNAE is given below.

Input: Local model updates $\hat{\mathrm{Up}}$
Output: Deviation detection Dv
Begin
Initialize $\hat{\mathrm{Up}}$, E^{coder}, W_{form}, B_{neck} and L_{pattern}
For 1 to each local model, do
#Encoder
Apply L2GNF

$$L_{\mathrm{GNF}} = \sqrt{\sum_{t=1}^{T} \hat{\mathrm{Up}}_t^{\,2}} \times \frac{\sum_{t=1}^{T} |\hat{\mathrm{Up}}_1 - \hat{\mathrm{Up}}_t|}{2T^2 \widetilde{M}}$$

Evaluate $E^{\mathrm{coder}} = W_{\mathrm{form}} \cdot (L_{\mathrm{GNF}})$
Execute bottleneck layer

$$B_{\mathrm{neck}}(\hat{\mathrm{Up}}) \xrightarrow{\text{capture}} L_{\mathrm{pattern}}$$

Implement decoder
Analyze deviation based on error
End For
Return Dv
End

If the deviation is detected, then the alert is sent to the hospital; otherwise, the local model gradients are updated into the global model.

Results and discussion

The proposed approach's performance is assessed by analogizing it with numerous existing studies. The proposed work is implemented on the PYTHON platform.

Dataset description

The proposed HD prediction framework is evaluated utilizing the HD Prediction Dataset (HDPD) from the Kaggle Database. The source link to access this dataset is demonstrated in Section "Conclusion". This dataset is utilized for training the HD prediction model that includes fundamental medical and demographic factors. Also, this dataset contains 1888 records, including 14 features that are important for predicting the heart attack and stroke risks. The 14 features are age (patient age), sex/ patient gender (male and female), cp/chest pain type (typical angina, atypical angina, non-anginal pain, and asymotomaic), trestbps/resting blood pressure, chol/ serum cholesterol level, fbs/fasting blood sugar (true and false), restecg/resting electrocardiographic results (normal, ST-T wave abnormality, and left ventricular hypertrophy), thalach/maximum heart rate achieved, exang/exercise-induced anginal (yes and no), oldpeak/ST depression (ST depression induced by excerice relative to rest), slope/peak exercise ST segement slope (upsloping and downsloping), ca/number of major vessels (0, 1, 2, and 3), thal/thalassemia types (normal, fixed defect, and reverisble defect), and target/hear attack risk (more chance of heart attack/ HD and less chance of heart attack/normal). Further, the collected data are preprocessed to ensure data integrity regarding duplicate data removal, missing value imputation, numeralization, and normalization. Moreover, to eliminate the class-imbalance issue, the SMOTE algorithm is applied, ensuring class-wise data balancing. Before databalancing, the dataset contains 977 and 911 data samples for HD and normal classes, correspondingly. After data balancing, the total number of data samples increased to 1954, containing 977 and 977 data samples for HD and normal classes, respectively. The detailed description of the class-wise dataset samples is depicted in Table 1.

From the whole data samples (1954), 80% (1563 samples) and 20% (391 samples) are allocated to perform training and testing, respectively.

Empirical assessment of the proposed work

To showcase the methodology's efficiency, the performance assessment is done.

Regarding Mean Square Error (MSE), Mean Absolute Error (MAE), and Normalized Mean Square Error (NMSE), the proposed L2-GNAE's performance is analogized with the prevailing AE, Variational AE (VAE), Denoising AE (DAE), and Sparse AE (SAE) in Fig. 3 and Table 2. Owing to the inclusion of L2GNF, the proposed L2-GNAE obtained a minimum Reconstruction Error (RE) (0.1024), L1 error (0.2173), MSE (0.1128), MAE (1.333), and NMSE (0.1443), indicating superior anomaly detection capability and reconstruction accuracy. However, the traditional methods exhibited comparatively higher error values due to their limited potential in effectively distinguishing between normal and anomalous model-update patterns. For instance, the traditional

Classes	Sample count (Before sampling)	Sample count (After sampling)
Heart disease	977	977
Normal	911	977
Total	1888	1954

Table 1. Dataset samples.

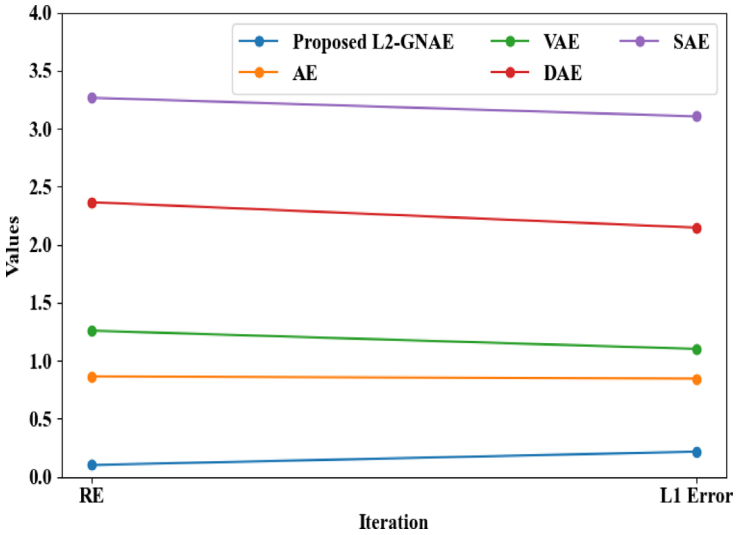


Fig. 3. Numerical investigation for model pattern deviation detection.

Techniques	MSE	MAE	NMSE
Proposed L2-GNAE	0.1128	1.333	0.1443
AE	0.8623	4.859	0.7583
VAE	1.0236	6.493	0.9482
DAE	2.1025	8.232	1.3358
SAE	3.2164	10.342	4.5893

Table 2. Robustness Analysis for model pattern deviation detection.

Methods	Accuracy (%)	Precision (%)	F-measure (%)	Specificity (%)	TPR (%)	TNR (%)	Training time (ms)
Proposed TGLRU	99.2145	99.0237	99.1046	99.2035	99.1492	99.1134	57,834
GRU	97.8452	97.6325	97.8412	97.8421	96.3894	97.9343	98,422
BiLSTM	94.6523	94.8712	94.3268	94.8417	92.5893	93.9353	108,493
LSTM	91.2549	91.3258	91.2417	91.2689	90.8943	91.7454	148,292
RNN	88.2147	88.2036	88.2149	88.7412	86.3432	87.5645	167,564

Table 3. Experimental investigation for HD prediction.

AE had a maximum RE, L1 error, MSE, MAE, and NMSE of 0.8652, 0.8471, 0.8623, 4.859, and 0.7583, respectively. Hence, the proposed method had impressive performance.

In Table 3 and Fig. 4, the proposed TGLRU's performance is weighed against the prevailing GRU, Long Short Term Memory (LSTM), Bidirectional LSTM (BiLSTM), and Recurrent Neural Network (RNN). Regarding a varying number of epochs, the proposed TGLRU obtained stable complexity owing to the TA function. Likewise, the proposed TGLRU attained accuracy, precision, f-measure, specificity, True Positive Rate (TPR), True Negative Rate (TNR), and training time of 99.2145%, 99.0237%, 99.1046%, 99.2035%, 99.1492%, 99.1134%, and 57834 ms, respectively. Yet, the existing approaches had considerable complexity and exhibited comparatively lower prediction performance owing to their limited potential in capturing complicated nonlinear relationships among clinical features. Therefore, by integrating Lipschitz regularization and the Triple Activation function with the traditional GRU, the superiority of the proposed method was evidenced, thereby eliminating overfitting and improving model generalization and learning efficiency.

In Table 4, the proposed SMCC's performance is contrasted with the prevailing ECC, Rivest-Shamir-Adleman (RSA), ElGamal, and Advanced Encryption Standard (AES), for data encryption. By utilizing the SMC, the proposed SMCC took encryption time of 4021 ms, decryption time of 4035 ms, and key generation time of 1189 ms, and attained a security level of 256bits and key size of 516bits. Traditional methods exhibited comparatively higher encryption time, decryption time, and key generation time, and a lower security level and key size. As a result, the traditional algorithms consumed an average encryption time of 11094 ms and decryption time of 10872 ms and obtained a security level of 148 bits. Therefore, the proposed method had high

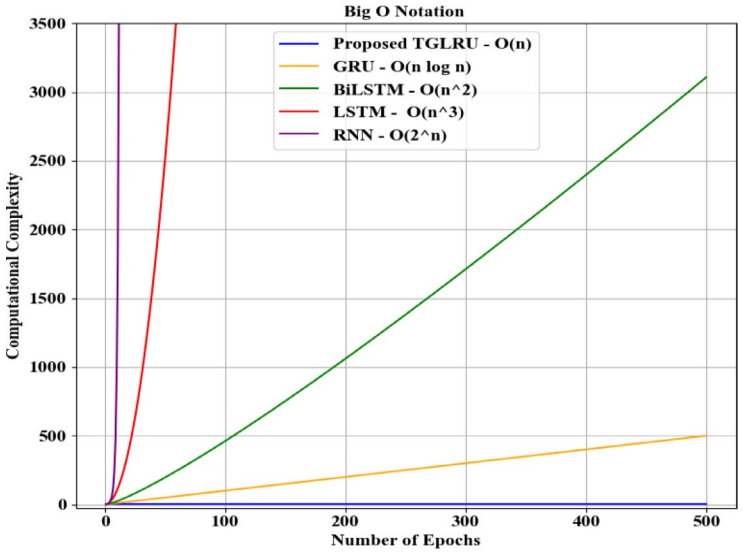


Fig. 4. Computational complexity analysis.

Methods	Encryption time (ms)	Decryption time (ms)	Security level (bits)	Key generation time (ms)	Key size (bits)
Proposed SMCC	4021	4035	256	1189	516
ECC	6359	6248	196	2894	384
RSA	9687	9652	156	3182	256
ElGamal	12,483	12,356	128	4895	196
AES	15,847	15,234	112	7242	192

Table 4. Empirical validation for data security.

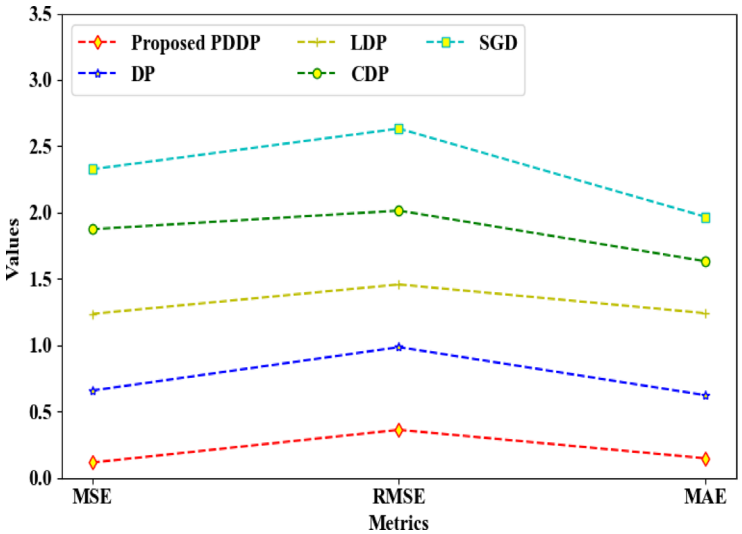


Fig. 5. Performance validation for model privacy.

robustness by integrating the SMC with the traditional ECC, enhancing cryptographic strength while reducing computational overhead and improving resistance against cryptographic attacks.

To prove the system's efficacy in model privacy, the performance analysis of the proposed PDDP and extant algorithms like PD, Local DP (LDP), Central DP (CDP), and Stochastic DP (SDP) is exhibited in Fig. 5 and Table 5. The proposed PDDP obtained Mean Squared Error (MSE), Root MSE (RMSE), Mean Absolute Error (MAE), NMSE, Mean Absolute Percentage Error (MAPE), and privacy budget of 0.1158, 0.3625, 0.1478, 0.893, 2.15%, and 0.03, respectively. This improved performance was due to the utilization of the Polynomial Decay Function

Techniques	MSE	RMSE	MAE	NMSE	MAPE (%)	Privacy budget
Proposed PDDP	0.1158	0.3625	0.1478	0.893	2.15	0.03
DP	0.6584	0.9856	0.6234	0.923	3.49	0.4
SDG	0.7647	1.0002	0.7093	1.129	4.61	0.7
K-Anonymity	0.8341	1.1978	0.9862	1.235	5.79	0.9
L-Diversity	0.9832	1.2364	1.1821	1.342	6.83	1.3
LDP	1.2365	1.4581	1.2415	1.492	7.31	1.6
CDP	1.8745	2.0147	1.6327	1.843	8.45	2.5
SGD	2.3269	2.6329	1.9682	2.503	9.42	3.1

Table 5. Numerical validation for model privacy.

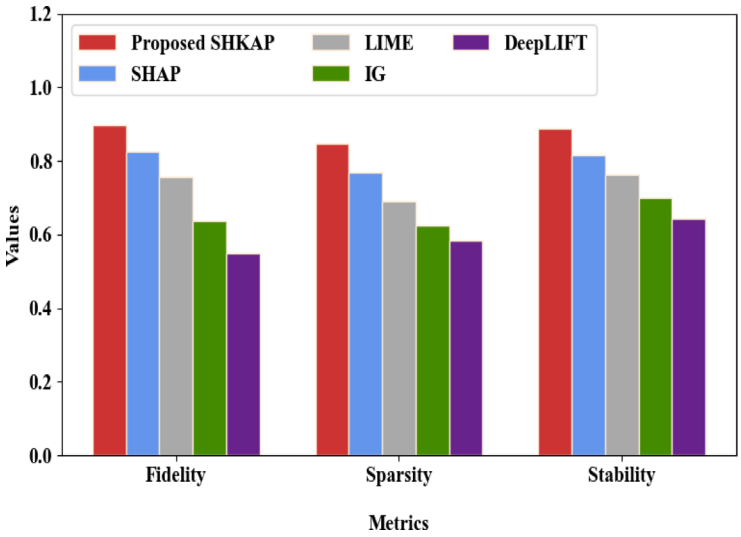


Fig. 6. Experimental assessment for deep explainability.

Techniques	Fidelity	Sparsity	Stability	Infidelity
Proposed SHKAP	0.8967	0.8479	0.8874	0.012
SHAP	0.8241	0.7684	0.8145	0.025
LIME	0.7564	0.6895	0.7632	0.061
IG	0.6359	0.6247	0.7012	0.072
DeepLIFT	0.5487	0.5843	0.6421	0.081

Table 6. Empirical analysis for deep explainability.

for adaptive privacy-budget allocation. Unlike conventional privacy-preserving methods that employed fixed privacy parameters, PDDP dynamically adjusted the privacy budget during model training. For instance, the traditional DP had maximum MSE (0.6584), RMSE (0.9856), MAE (0.6234), NMSE (0.923), MAPE (3.49%), and privacy budget (0.4). By determining the privacy budget appropriately, the proposed method achieved better outcomes.

The proposed SHKAP's performance is analogized with the extant SHAP, Local Interpretable Model-agnostic Explanations (LIME), Integrated Gradient (IG), and Deep Learning Important FeaTures (DeepLIFT) in Fig. 6 and Table 6. Here, the proposed SHKAP integrated Kaniadakis Entropy with conventional SHAP to improve feature-weight assignment and explanation quality. As a result, the proposed SHKAP attained a fidelity of 0.8967, sparsity of 0.8479, stability of 0.8874, and infidelity of 0.012, indicating robust generation of explanations and maintaining consistency across different samples and perturbations. In contrast, the conventional methods struggled to generate concise explanations for the predicted outcome due to their inability to accurately estimate feature importance and handle complex nonlinear relationships among clinical attributes. For instance, the conventional SHAP obtained a fidelity of 0.8241, stability of 0.8145, sparsity of 0.7684, and infedility of 0.025. Overall, the proposed approach had significant performance.

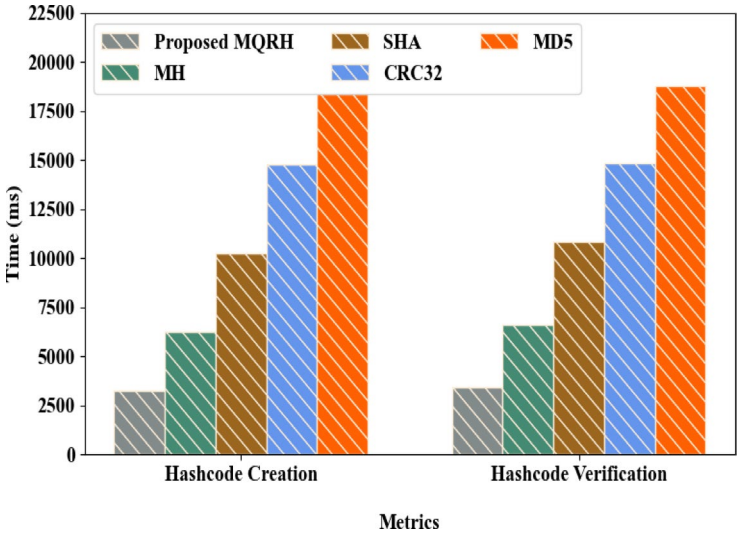


Fig. 7. Numerical analysis for hash generation.

Techniques	Hashcode creation Time (ms)	Hashcode verification time (ms)	Collision rate (%)
Proposed MQRH	3256	3418	0.02
MH	6241	6584	0.35
SHA-512	10,239	10,842	0.79
CRC32	14,758	14,853	1.26
MD5	18,326	18,754	2.11

Table 7. Effectiveness analysis for hash generation.

Authors' name	Algorithms	Accuracy (%)	Precision (%)	F-measure (%)
Proposed work	TGLRU	99.2145	99.0237	99.1046
[26]	Graph Neural Network (GNN)	98.77	96.04	96.52
[27]	TabNet	82.2	–	–
[28]	Homomorphic Encryption-Driven Logistic Regression (HELR) framework	84.00	–	–
[29]	Ensemble machine learning	83.21	84.09	86.13
[30]	soft-margin L1-regularised Support Vector Machine (sSVM)	98.57	–	–
[31,32]	CatBoost + Transformer	95.12	–	95.24
[33]	Federated Learning with Weighted Conglomeration Optimization (FLWCO)	97.27	95.28	92.46

Table 8. Comparative assessment of the proposed work.

In Fig. 7 and Table 7, the proposed MQRH's efficiency is validated by analogizing it with prevailing MH, Secure Hashing Algorithm-512 (SHA-512), Cyclic Redundancy Check-32 bit (CRC32), and Message Digest-5 (MD5). From these results, it was proven that the traditional methods showed comparatively lower performance in hash generation with high hashcode creation time, hashcode verification time, and collision rate. For instance, the traditional MH achieved 6241 ms hashcode creation time, 6584 ms hashcode verification time, and 0.35% collision rate due to the improper seed value selection. In contrast, by performing Quadratic Residue Equation-based optimal seed value selection, the proposed MQRH took 3256 ms and 3418 ms to perform hashcode creation and hashcode verification, respectively, and achieved a much lower collision rate (0.02%). Therefore, the proposed method obtained high robustness.

Comparative validation

The comparative investigation of the proposed and existing studies is done.

Table 8 showcases the comparative assessment of the proposed work and existing studies. The proposed TGLRU obtained accuracy, precision, and f-measure of 99.2145%, 99.0237%, and 99.1046%, respectively, owing to the LR technique and TA function. Similarly, the traditional studies established techniques, such as sSVM, TabNet, GNN, and ensemble machine learning, to perform privacy-aware HD prediction. The traditional GNN[26]

had a minimum accuracy (98.77%), precision (96.04%), and f-measure (96.52%) due to the vanishing gradient issues. Similarly, Rahman et al.[31,32] and Dash et al.[33] demonstrated comparatively lower f-measure of 95.24% and 92.46% owing to the limited ability to effectively capture long-range temporal dependencies and complex nonlinear relationships in heterogeneous data. Additionally, the HELR-based HD prediction model exhibited relatively lower accuracy (84%) than the proposed framework due to its inability to handle high-dimensional medical data. Thus, the proposed methodology was more effective than conventional studies.

Discussion

To showcase the model's superiority in a privacy-aware healthcare framework with model pattern-deviation detection for heart-disease prediction, an empirical assessment and comparative analysis were conducted. The performance of the proposed approaches, such as data encryption and decryption (SMCC), hash code generation (MQRH), Model Privacy (PDDP), shared model pattern deviation detection (L2-GNAE), disease classification (TGLRU), and deep explainability (SHKAP), was validated by considering several quantitative and qualitative metrics. The experimental results demonstrated that the proposed SMCC effectively secured sensitive healthcare information by achieving a higher security level (256bits) while maintaining lower encryption time (4021 ms) and decryption time (4035 ms). The utilization of the Siegel Modular Curve improved cryptographic robustness and enhanced resistance against potential security threats, making it suitable for privacy-sensitive healthcare environments. Further, the data integrity was enhanced using the proposed MQRH, which performed hash-code generation and verification by using the Quadratic Residue Equation. This integration enhanced the randomness of seed-value initialization and reduced the possibility of hash collisions, resulting in effective detection of unauthorized modification in healthcare records with 3256 ms hash code creation time and 3418 ms hash code verification time. Then, the privacy-preserving capability of the proposed framework was validated using the proposed PDDP. Unlike conventional DP approaches, PDDP dynamically adjusted the privacy budget using a polynomial decay strategy, effectively balancing privacy protection and predictive utility. Also, the proposed PDDP reduced information leakage risks during collaborative model training and sharing by attaining very low MSE, RMSE, and MAE values of 0.1158, 0.3625, and 0.1478, respectively. Furthermore, the proposed L2-GNAE was applied to detect deviations in shared model patterns within the federated learning environment. Existing FL frameworks focused on model aggregation and overlooked the identification of abnormal model updates generated by compromised participants. The proposed L2-GNAE effectively addressed this limitation by integrating the L2 Gini Norm filtering with the conventional autoencoders. This integration effectively distinguished normal and abnormal model updates by preventing the propagation of malicious or corrupted model parameters with lower RE (0.1024), L1 Error (0.2173), and MSE (0.1128). Moreover, for disease prediction, the proposed TGLRU illustrated better classification performance when analogized with traditional DL models. The integration of Lipschitz regularization improved model generalization and mitigated overfitting, while the triple activation mechanism enhanced learning efficiency and feature representation capability. These integrations helped to accurately capture complex nonlinear relationships among clinical features and provided reliable HD prediction with reduced computational complexity and improved accuracy (99.2145%), precision (99.0237%), and f-measure (99.1046%). Additionally, the proposed SKHAP provided transparent and explainable prediction by using the Kaniadakis Entropy, which assisted in generating more reliable feature-attribution scores and improved explanation quality. Also, the proposed SKHAP offered consistent and meaningful explanations by attaining higher fidelity (0.8967), stability (0.8874), and sparsity (0.8479). Likewise, the comparative evaluation was done by comparing the proposed framework's effectiveness with traditional works. From the evaluation, it was proven that the novel TGLRU accurately detected HD diseases with improved accuracy, precision, and f-measure value than the traditional works. The proposed framework showed superior performance than the conventional works due to the integration of data encryption and decryption, hash code generation and verification, shared model pattern deviation detection, model privacy, HD classification, and deep explainability within the FL model. Consequently, the proposed privacy-aware healthcare framework provided a comprehensive, secure, reliable, and trustworthy solution for HD prediction in modern healthcare environments.

Limitation

Although the proposed framework achieves superior performance in privacy-aware HD prediction, it introduces several limitations. Initially, the proposed work integrates multiple components to improve the methodology's security, privacy, reliability, and interpretability. Although these components improve the overall performance of the proposed work during FL and privacy-aware HD prediction, they also present additional computational as well as communication overhead. Additionally, the proposed work uses a fixed privacy-budget adaptation strategy to regulate the privacy budget during federated learning. Although this mechanism effectively balances privacy and model utility, the privacy-budget adaptation remains fixed once the decay parameters are selected. This limits the proposed work's ability to respond optimally to dynamic healthcare environments. Further, the proposed work experience increased latency when deployed in large-scale federated healthcare systems, which include a substantial number of participating hospitals, medical centers, or edge devices. These factors can significantly affect real-time deployment in resource-constrained or large-scale healthcare environments.

Conclusion

Here, the paper proposed a privacy-aware healthcare framework with model pattern-deviation detection for HD prediction using L2-GNAE and PDDP. The proposed L2-GNAE was generalized well enough to detect the deviations in the shared model patterns, improving the healthcare reliability by attaining 0.1024 RE, 0.2173 L1 Error, and 0.1128 MSE. To ensure model privacy, the proposed PDDP was established. The proposed PDDP comparatively showed robust performance by enhancing the model privacy with 0.3625 RMSE, 0.1478 MAE,

and 0.03 privacy budget. The data encryption was done using SMCC to protect the SHD, attaining lower encryption time (4021 ms) and higher security level (256bits) and key size (516bits). Furthermore, the proposed MQRH effectively mitigated the insider threats with a minimum hash code verification time and collision rate of 3418 ms and 0.02%, correspondingly. Also, the proposed TGLRU significantly predicted the HD with an accuracy of 99.2145%. Therefore, the proposed framework performed better than the traditional works by offering a reliable, trustworthy, and privacy-aware solution for intelligent HD prediction in federated healthcare environments, thereby achieving superior security.

Practical implications

The proposed framework enables secure, reliable, and privacy-preserving HD prediction in federated healthcare environments. Additionally, the proposed L2-GNAE aids in identifying the SMPD in real time, improving the reliability of clinical decision-making and treatment planning. Also, TGLRU supports early HD diagnosis, while SKHAP enhances the transparency of prediction outcomes. Overall, the proposed framework is highly appropriate for deployment in smart hospitals, telemedicine platforms, remote patient-monitoring systems, and federated healthcare networks.

Future scope

The work will focus on optimizing the privacy budget dynamically using reinforcement learning to increase the model's resilience in the future. Similarly, to reduce latency in emergency situations, edge/fog computing will be integrated. Furthermore, lightweight model optimization and efficient parameter-sharing mechanisms will be developed to reduce computational complexity and improve scalability in large-scale federated healthcare environments.

Data availability

(https://www.kaggle.com/datasets/mfarhaannazirkhan/heart-dataset.

Received: 5 March 2026; Accepted: 29 June 2026
Published online: 08 July 2026

References

1. Almutairi, S. et al. A context-aware MRIPPER algorithm for heart disease prediction. *J. Healthc. Eng.* **2022**, 1–11. https://doi.org/10.1155/2022/7853604 (2022).
2. Kathole, A. B. et al. Electronic health records protection strategy by using blockchain approach. *Multimedia Tools and Applications* **83**, 86883–86894. https://doi.org/10.1007/s11042-024-19772-x (2024).
3. Pande, S. D. et al. A novel chatbot driven by sentiment analysis using a capsule network and BiLSTM for online user feedback on consumer electronics. *IEEE Trans. Consum. Electron.* **71**(3), 7933–7946. https://doi.org/10.1109/TCE.2025.3593110 (2025).
4. Patil, S. D., Kathole, A. B., Kumbhare, S., Vhatkar, K. & Kimbahune, V. V. A blockchain-based approach to ensuring the security of electronic Data. *Int J Intell Syst Appl Eng* **12**(11), 649–655 (2024).
5. Abaoud, M., Almuqrin, M. A. & Khan, M. F. Advancing federated learning through novel mechanism for privacy preservation in healthcare applications. *IEEE Access* **11**, 83562–83579. https://doi.org/10.1109/ACCESS.2023.3301162 (2023).
6. Otoum, Y., & Nayak, A. (2025). Differential privacy-driven framework for enhancing heart disease prediction. In *ICC 2025 - IEEE International Conference on Communications*, 1–7. https://doi.org/10.1109/ICC52391.2025.11161731.
7. Singh, J. P., Aqsa, A., Ghani, I., Sonani, R. & Govindarajan, V. Privacy-aware hierarchical federated learning in healthcare: Integrating differential privacy and secure multi-party computation. *Future Internet* **17**, 1–24. https://doi.org/10.3390/fi17080345 (2025).
8. Batchala, R., Jain, P., Gyanchandani, M., Shukla, S. & Wadhvani, R. Enhance differential privacy mechanisms for clinical data analysis using CNNs and reinforcement learning. *J. Big Data* **12**, 1–18. https://doi.org/10.1186/s40537-025-01215-5 (2025).
9. Owusu-agyemeng, K. et al. MSDP: Multi-scheme privacy-preserving deep learning via differential privacy. *Pers. Ubiquitous Comput.* **27**, 221–233. https://doi.org/10.1007/s00779-021-01545-0 (2023).
10. Kathole, A. B. et al. Energy-aware UAV based on blockchain model using IoE application in 6G network-driven cybertwin. *Energies* **15**, 1–17. https://doi.org/10.3390/en15218304 (2022).
11. Kumbhare, S., Kathole, A. B. & Shinde, S. Federated learning aided breast cancer detection with intelligent heuristic-based deep learning framework. *Biomed. Signal Process. Control* **86**, 1–18. https://doi.org/10.1016/j.bspc.2023.105080 (2023).
12. Kathole, A., Patil, S., Jadhav, D., Pathak, H. & Mirge, A. S. Development of student intent-based educational chatbot system with adaptive and attentive DTCN on symmetric convolution approach. *MethodsX* **15**, 103542. https://doi.org/10.1016/j.mex.2025.103542 (2025).
13. Wan, S., Wan, F. & Dai, X. Machine learning approaches for cardiovascular disease prediction: A review. *Arch Cardiovasc Dis* https://doi.org/10.1016/j.acvd.2025.04.055 (2025).
14. Harjani, M., Grover, M., Sharma, N., & Kaushik, I. Analysis of various machine learning algorithm for cardiac pulse prediction. In *2019 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS)*, 244–249. https://doi.org/10.1109/ICCCIS48478.2019.8974519 (2019).
15. Sharma, A., Sharma, N., Kaushik, I., Kumar, S., & Khatoon, N. Predictive analysis of type 2 diabetes using hybrid ML model and IoT. In *IoT Security Paradigms and Applications*, 303–320. https://doi.org/10.1201/9781003054115-14 (2020).
16. Tyagi, N., Bhushan, B., Gautam, S., Sharma, N., & Kumar, S. An Integrated Approach of Blockchain & Big Data in Health Care Sector. In *Applications of Machine Learning in Big-Data Analytics and Cloud Computing*, 183–205. https://www.taylorfrancis.com/chapters/edit/10.1201/9781003337218-9/integrated-approach-blockchain-big-data-health-care-sector-nitin-tyagi-bharat-bhushan-siddharth-gautam-nikhil-sharma-santosh-kumar (2022).
17. Sharma, N. & Shambharkar, P. G. Towards secure healthcare: SA-GBO-ODBN model utilizing blockchain and deep learning for data handling and diagnosis. *Comput. J.* **68**(10), 1386–1423. https://doi.org/10.1093/comjnl/bxaf045 (2025).
18. Torfi, A., Fox, E. A. & Reddy, C. K. Differentially private synthetic medical data generation using convolutional GANs. *Inf. Sci.* **586**, 485–500. https://doi.org/10.1016/j.ins.2021.12.018 (2022).
19. Pandey, S., Singh, O., Pandey, A. & Pandey, C. Robust and privacy-preserving federated learning against malicious clients: A Bulyan-based adaptive differential privacy framework. *IEEE Access* **13**, 139931–139943. https://doi.org/10.1109/ACCESS.2025.3596627 (2025).
20. Lian, Z. et al. DEEP-FEL: Decentralized, efficient and privacy-enhanced federated edge learning for healthcare cyber physical systems. *IEEE Trans. Netw. Sci. Eng.* **9**(5), 1–12. https://doi.org/10.1109/TNSE.2022.3175945 (2022).

21. Fang, M. L., Dhami, D. S., & Kersting, K. DP-CTGAN : Differentially Private Medical Data Generation using CTGANs. In *International Conference on Artificial Intelligence in Medicine*, 1–10. https://www.aiml.informatik.tu-darmstadt.de/papers/fang2022dpctgan.pdf (2022).
22. Tian, Y. et al. Robust and privacy-preserving decentralized deep federated learning training: Focusing on digital healthcare applications. *IEEE/ACM Trans. Comput. Biol. Bioinform.* **21**(4), 890–901. https://doi.org/10.1109/TCBB.2023.3243932 (2024).
23. Alammar, S., Aldawsari, H., Alsahly, A., Alghamdi, K. A. & Khan, M. A. Federated explainable AI for privacy-preserving cardiovascular disease detection using wearable devices. *IEEE Trans. Consum. Electron.* **72**(1), 2164–2175. https://doi.org/10.1109/TCE.2025.3647850 (2026).
24. Shajimon, S., Shukla, R. M. & Patra, A. N. Precision at heart: An IoT-based vertical federated learning approach for heterogeneous data-driven cardiovascular disease risk prediction. *Comput. Methods Programs Biomed.* **273**, 1–11. https://doi.org/10.1016/j.cmpb.2025.109079 (2026).
25. Naveenkumar, N., Subarna, T. G., Pragadeeswaran, S. & Karuppuchamy, V. Federated learning based hybrid Dlbased transformer model for heart disease detection with explainable AI. *Biomed. Signal Process. Control* **120**, 1–16. https://doi.org/10.1016/j.bspc.2026.110076 (2026).
26. Noor, M. A., Hassan, S. B., Alam, S. B., Lameesa, A. & Siddique, A. R. A privacy-preserving federated learning framework with graph neural networks for enhanced heart attack risk prediction. *Array* **27**, 1–14. https://doi.org/10.1016/j.array.2025.100500 (2025).
27. Otoum, Y., Hu, C., Said, E. H. & Nayak, A. Enhancing heart disease prediction with federated learning and blockchain integration. *Future Internet* **16**, 1–17. https://doi.org/10.3390/fi16100372 (2024).
28. Naresh, V. S. & Reddi, S. Exploring the future of privacy - preserving heart disease prediction: A fully homomorphic encryption - driven logistic regression approach. *J. Big Data* **12**, 1–27. https://doi.org/10.1186/s40537-025-01098-6 (2025).
29. Azar, A. et al. FMLCA : explainable and privacy-preserving federated machine learning classification algorithms for predicting heart disease in patients. *Eur. J. Med. Res.* https://doi.org/10.1186/s40001-026-04023-6 (2026).
30. Bebortta, S., Tripathy, S. S., Basheer, S. & Chowdhary, C. L. FedEHR: A federated learning approach towards the prediction of heart diseases in IoT-based electronic health records. *Diagnostics* **13**, 1–19. https://doi.org/10.3390/diagnostics13203166 (2023).
31. Rahman, A. U. R. et al. DP-FedHybrid: A differentially private federated stacking framework integrating tree-based and transformer models for secure heart disease prediction. *IEEE Access* **14**, 64681–64695. https://doi.org/10.1109/ACCESS.2026.3687059 (2026).
32. Rahman, W., Khan, R. & Nijim, M. Privacy-preserving deep learning for disease diagnosis in medical imaging : A systematic review. *IEEE Access* **14**, 7248–7269. https://doi.org/10.1109/ACCESS.2025.3647561 (2026).
33. Dash, S. et al. Privacy-preserving diabetes and heart disease prediction via federated learning and WCO. *Int. J. Comput. Intell. Syst.* **18**, 1–26. https://doi.org/10.1007/s44196-025-00956-8 (2025).

Acknowledgements

The researchers would like to thank Chosun University, South Korea for financial support.

Author contributions

Formal Analysis-Basant Kumar, Rashmi Dwivedi; Resources, Design and coding : Rashmi Dwivedi,Vivek Mishra; Wrting-Original Draft—Vidushi; Writing—Rashmi Dwivedi; Review and Editing—SeongKi Kim.

Funding

This study was supported by research funds from Chosun University, 2025.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-60544-4.

Correspondence and requests for materials should be addressed to S.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

© The Author(s) 2026